PROBLEM STATEMENT

Smart Traffic Signal Scheduling System Using Circular Linked List

1. Introduction

Traffic congestion at road intersections can cause delays, fuel wastage, and confusion when signal timings are not managed in an organized way. A simple scheduling system can represent each traffic signal as a node and cycle through the signals in a fixed order.

2. Problem Definition

Design and implement a Smart Traffic Signal Scheduling System using a circular linked list. Each node represents a road direction or lane group and stores its signal name and green-light duration. The last node must point back to the first node so that signal scheduling continues repeatedly in a circular manner.

3. Objectives

- Represent each traffic signal using a node in a circular linked list.
- Add and maintain multiple signals in a repeating sequence.
- Display the active signal and its scheduled green-light duration.
- Move to the next signal after the current duration ends.
- Continue cycling through all signals until the system is stopped.

4. Scope and Assumptions

This is a data-structures mini project that demonstrates scheduling logic. Signal changes may be simulated through program output and a timer/delay. It does not connect to real traffic lights, sensors, cameras, or emergency-vehicle detection. For simplicity, each signal uses a predefined duration.

5. Expected Outcome

The program should create a circular linked list of traffic signals, show the currently active signal, wait for its configured duration (or simulate the wait), and advance to the next signal. After the last signal, it should return to the first signal without reaching a NULL link.

6. Data Structure Used

Circular Linked List: A linked list in which the next pointer of the final node points to the first node. This structure is suitable for repeating schedules because traversal can continue cyclically.